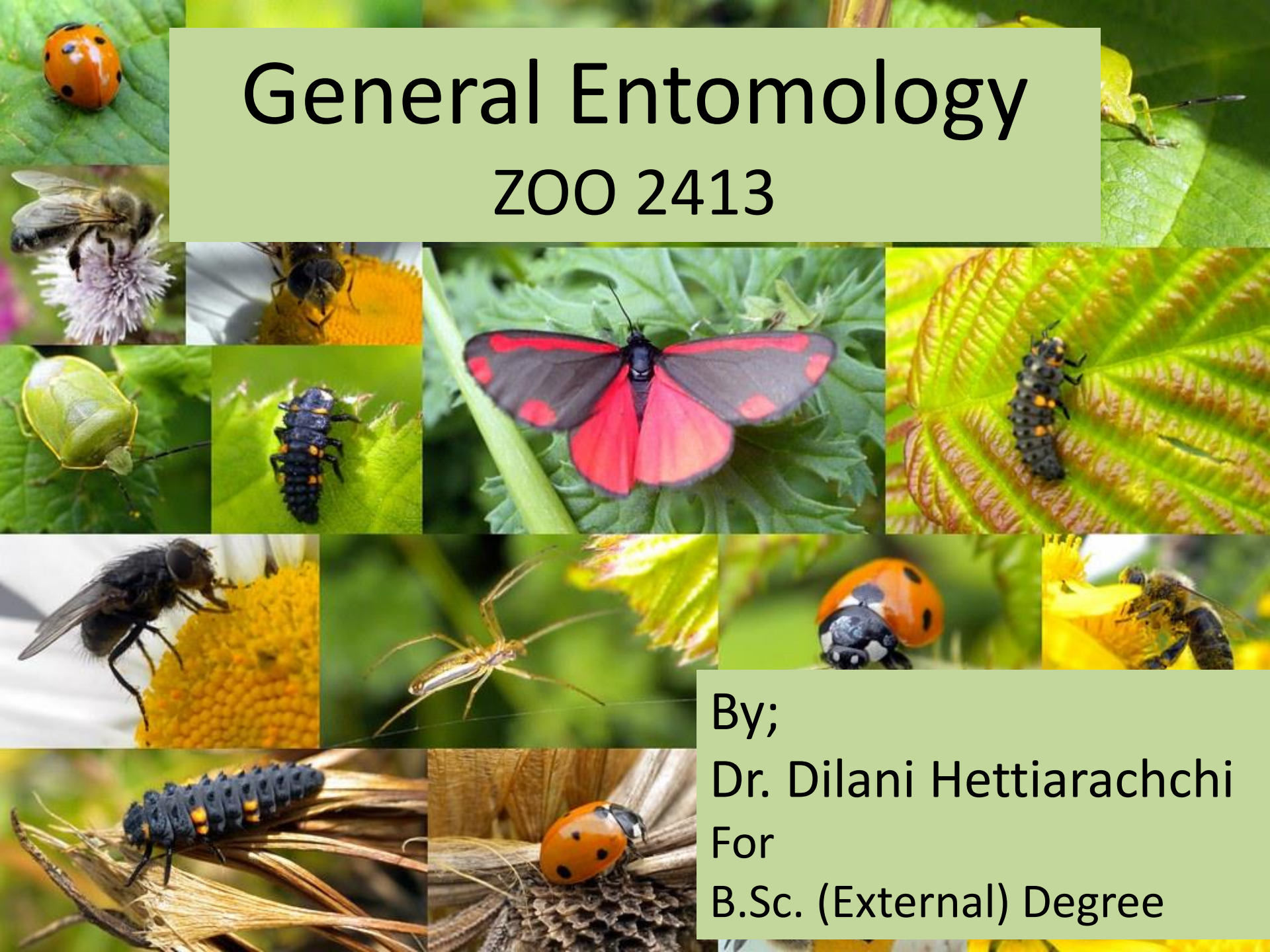


General Entomology

ZOO 2413



By;
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For
B.Sc. (External) Degree

Lesson Objectives

After completion you should be able to;

- Describe the structure and components of the insect integument
- Explain the body regions of insect integument/exoskeleton
- Explain the process of ecdysis

External structures of insect body and their functions

The Integument

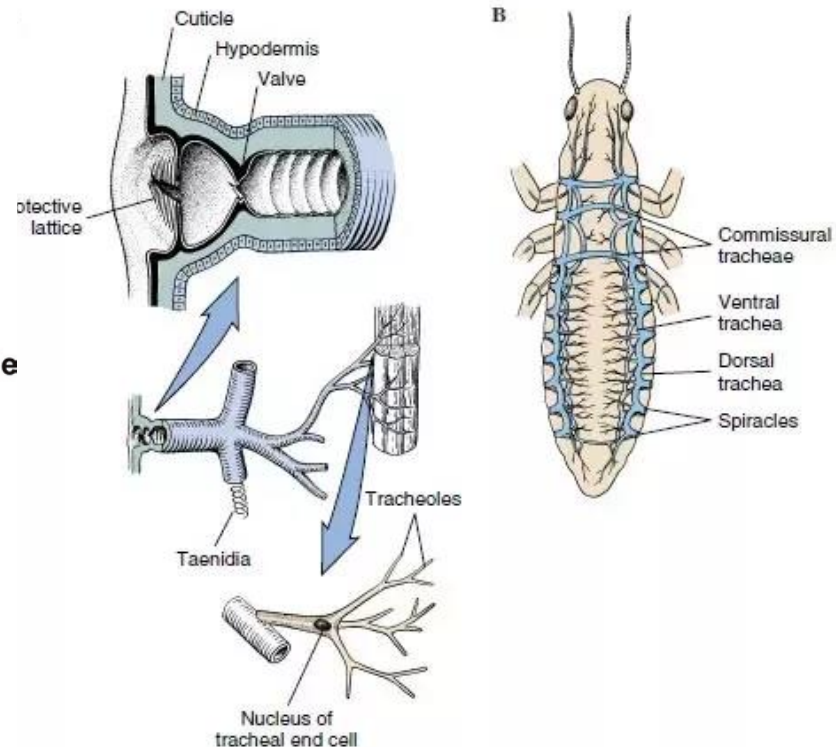
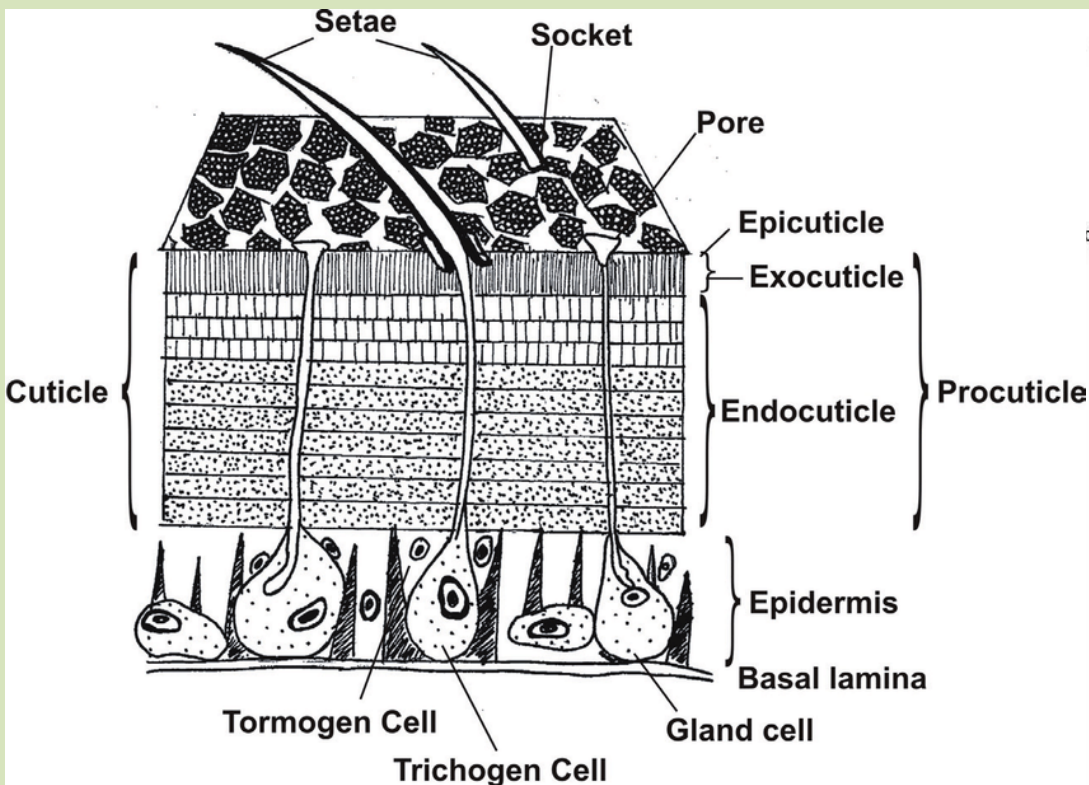
- Integument = body covering
- Insect integument can be divided into three parts;
 1. The Cuticle – non-cellular layer
 2. The epidermis – cellular layer
 3. The basement membrane - Non- cellular layer

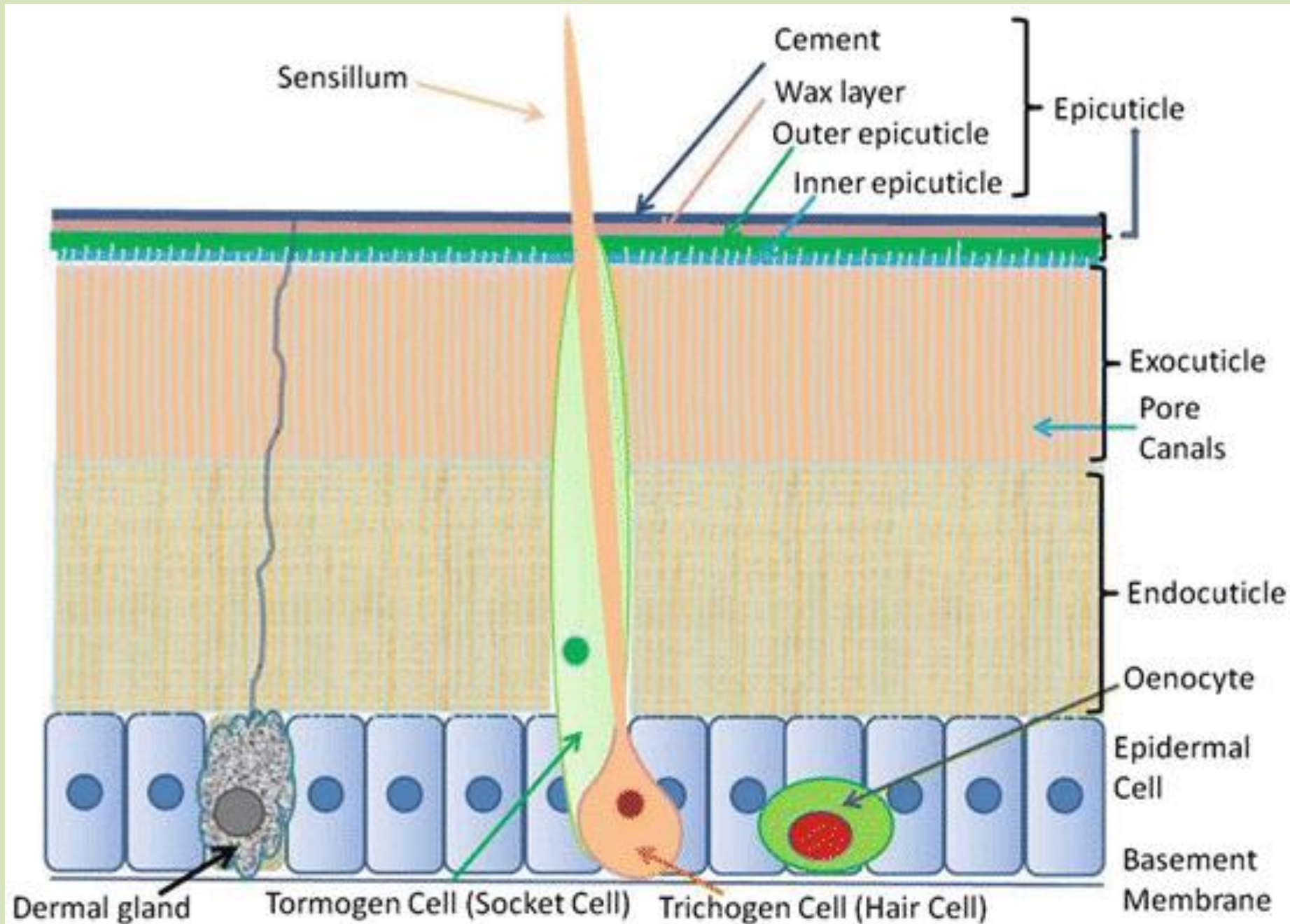
Cuticle

- Comprises up to one-half the dry weight in certain species
- complex, non-cellular layer
- Secrete largely by the epidermis
- It forms the outermost investment of the body and its appendages
- Invaginate to form endoskeleton structures

Cuticle cont.

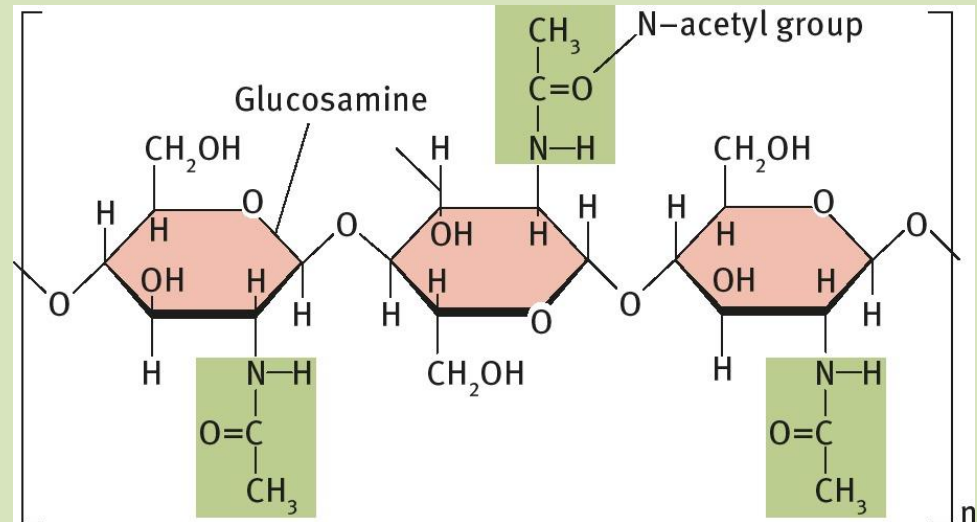
- It also provide the lining of the;
 - Tracheal system
 - Some glands
 - Parts of the alimentary canal
 - Parts of the reproductive tract





Cuticle cont.

- Two major components of the cuticle are
 - carbohydrate chitin
- ✓ 25% - 60% of the dry weight
- ✓ insoluble in water, alkalis, dilute acids and organic solvents
- ✓ Dissolve in concentrated mineral acids and sodium hypochlorite
- ✓ Tannish colour



Cuticle cont.

→ proteins

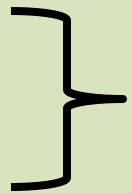
- ✓ Sometimes collectively known as arthropodin
- ✓ There are more than dozen protein components identified

- Other components are;

→ Phenolic precursors (quinones) – helps to link amino acid chains

→ waxes

→ Other lipids

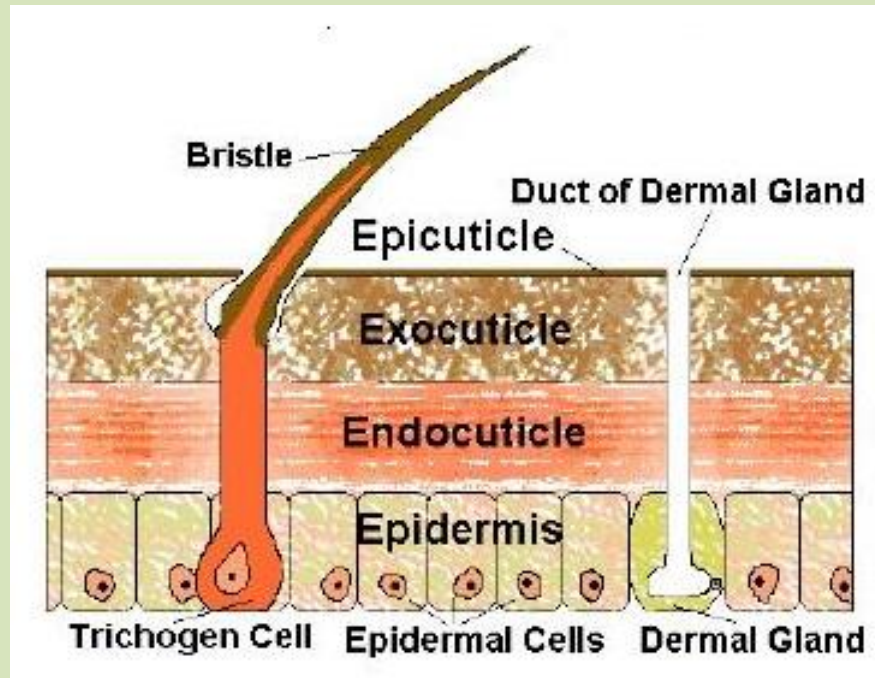


Secreted by oenocytes

→ Enzymes – responsible for the synthesis and degradation of major cuticular materials

Cuticle cont.

- Two main layers;
 - Outer epicuticle
 - Inner procuticle
 - exocuticle
 - endocuticle



Epicuticle

- Very thin layer (0.03 – 4.0 μm)
- Devoid of chitin and complex chemical composition
- Contain lipids
- unpigmented
- Resistant to acids and organic solvents
- Before ecdysis the epicuticle is folded

Epicuticle cont.

- Have at least four layers depending on the insect species and life cycle stages
1. Wax layer
 - Consist of both lipids and hydrocarbons
 - Can function as kairomones, recognition cues, and sex attractants
 - There are tiny wax-canals continuous with pore-canals

Epicuticle cont.

2. Cement layer

- secreted by epidermal glands
- determine the surface properties of the cuticle (ex: hydrophilic, hydrophobic nature)
- Protective barrier
- In some insects this layer maybe absent

3. Outer epicuticle (cuticulin)

4. Inner epicuticle

Procuticle

- Secreted by epidermal cells
- Thicker than the epicuticle (up to 200 μm)
- Make up the bulk of the integument

Exocuticle

- Outer layer
- Composed of a homogeneous dense matrix
- Harden through sclerotization
- Often pigmented

Exocuticle cont.

- Highly stabilized layer of the procuticle that is resistant to the action of moulting fluid
- Shed entirely as the exuviae during ecdysis
- Typically thin in soft-bodied insects and thicker in hard-bodied insects

endocuticle

- Softer
- Capable of increasing considerably through post-ecdysial deposition
- Have protein and chitin fibers
- Seen banded composed of light and dark layers
- Banded in structure where the darker layers are those deposited during the day and lighter layers during night

Endocuticle cont.

- Have numerous pore-canal (1 micrometer or less)
- Connecting tubes between the cellular and the cuticular layers
- These pore-canal may transport materials to outer procuticle and epicuticle
- Endocuticle can be digested by moulting fluid

Sclerotization

- Chitin is not the reason for the hardness of the cuticle
- Hardening is due to tanning of protein
- This happen after **eclosion** (emergence of an immature stage from the egg) or **ecdysis** (shedding of the cuticle) or subsequent **expansion** of an insect
- Adjacent protein chains are cross-linked by *o*-quinones.
- Exocuticle become hard, tough, inelastic

The Epidermis

- Continouse, single layer of cells
- Secrete the cuticle
- Produce the moulting fluid
- Scattered among the normal epithelial cells are gland cells

The Basement membrane

- Continuous amorphous layer (0.5 μm in thickness)
- Composed of mucopolysaccharide

Functions of integument

- Protects internal delicate organs from mechanical injury.
- Provide a framework for insect body
- Protect body from excessive moisture and drying
- Protect the body from attack of many diseases
- In aquatic insects respiratory gases exchange through cuticle.
- Excellent structure for attaching a musculature system with good leverage

Coloration

- Colours of adults and immature insets can be grouped into three classes;
 1. Pigmentary/chemical colours
 2. structural/physical colours
 3. combination or chemico-physical colours



1. Pigmentary/chemical colours

- Due to various pigments present in the cuticle, scales on the cuticle, epidermal cells, and fat-body, etc.
- Due to chemical substances
- They absorb some wavelengths from the light and reflect others
- Produced by complex mixture of different pigments

- Same colour maybe achieved in different ways
- Examples of insects pigments
 - Melanins – yellow, brown, black
 - Carotenoids – red and yellow
 - [Derived from plant tissues]
 - Pterins – red, yellow, white
 - Ommochromes – red, yellow, brown
 - [commonly found as eye pigment]
 - Anthraquinones – red, orange
 - [common in insect scales]
 - Aphins – red, orange, yellow
 - [confined to aphids]

- Chlorophyll derivatives – greenish
- hemoglobin derivatives – reddish
- Anthoxanthins – whitish, yellow
- Insectoverdins – green
- Flavines – greenish yellow

2. Structural/physical colours

- Due to the effects of various structural configurations on light
- Colours results from the reflection of light in all directions by irregular surfaces or grsnnules beneath the surfaces

Ex – White in several butterflies are produced by the light-scattering effect of the irregular surfaces of scale



3. combination or chemico-physical colours

- Structural modification in conjunction with a layer of pigments
- Much commoner than purely structural colours

Ex; In the butterfly *Teracolus* a red pigment in the scale wall combines with a structural violet to produce magenta

Cuticular appendages

1. Setae

- Commonly known as hairs
- Each arise from a cup-like pit or alveolus
- Hollow structures
- Extensions of the exocuticle
- Each is produced by a single, usually enlarged **trichogen cell**
- Main kinds of setae and other modifications are as follows;
 - a) Clothing hair
 - plumose hair
 - bristle

b) Scales (Lepidoptera, Collembola, some Thysanura, Coleoptera, Hymenoptera and Diptera

c) Glandular setae

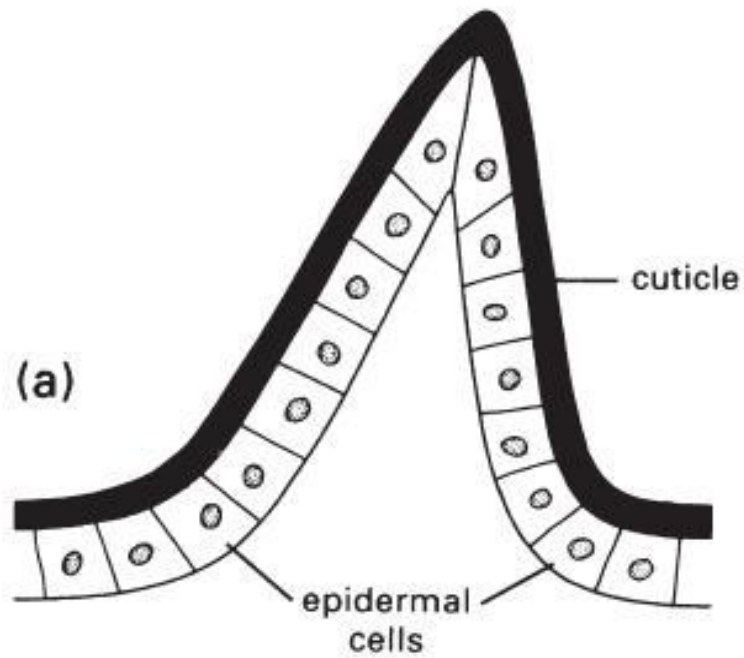
d) Sensory setae



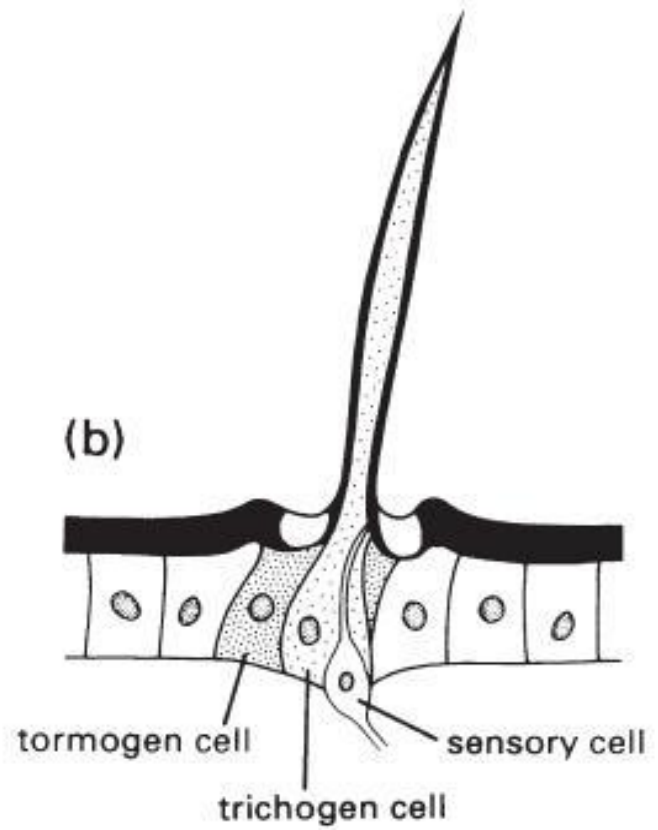
2. Spurs

- occur on the legs
- Differ from setae → multicellular origin





Spine



Setae

Cuticular Processes

Spines

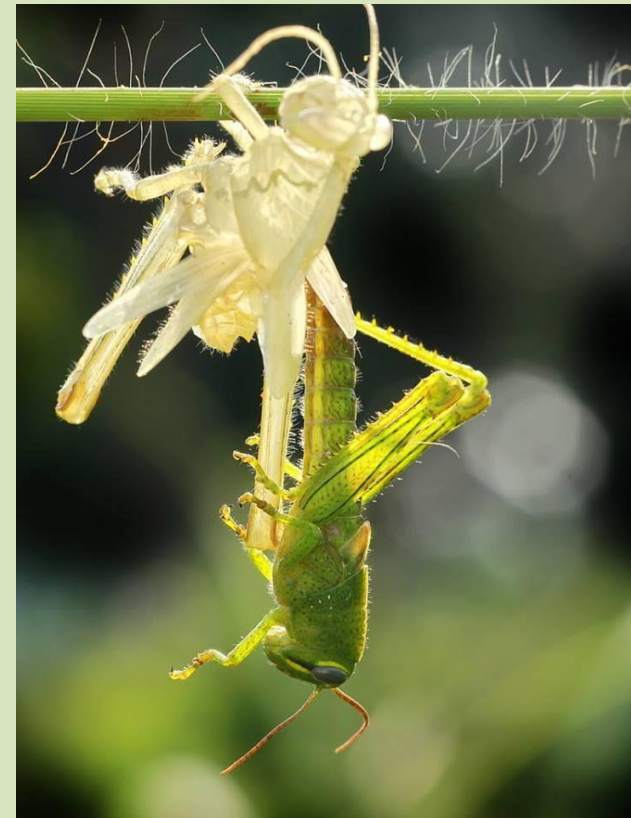
- Outgrowth of the cuticle
- More or less thorn – like
- differ from spine like setae → produced by undifferentiated epidermal cells and usually multicellular in origin (not always)

Nodules

Horns (certain male Coleopterans)

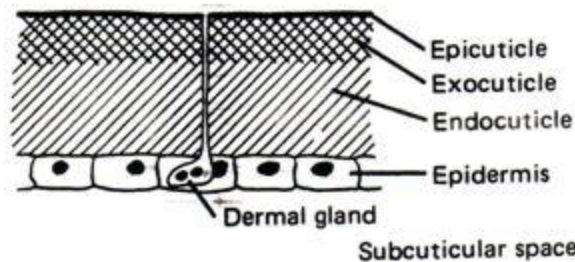
Molting/ Ecdysis

- Periodic shedding of cuticle followed by formation of a new cuticle
- Mechanism facilitating growth
- Under hormonal control (hormone – ecdysone)

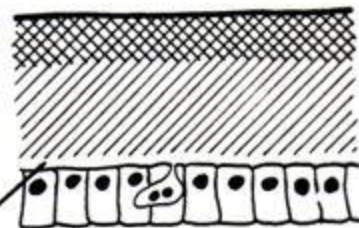


The seven basic phases of insect molting.

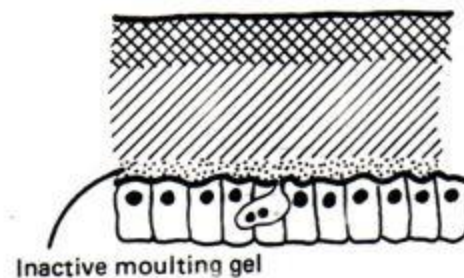
MATURE CUTICLE



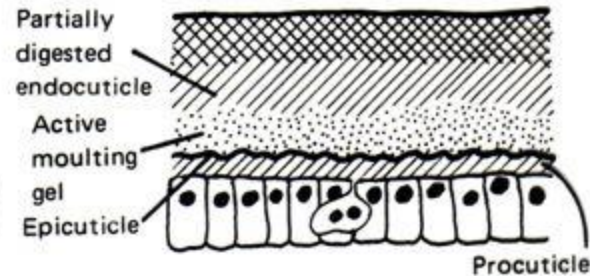
1. APOLYSIS



2. NEW EPICUTICLE PRODUCED



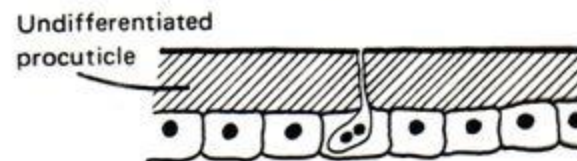
3. PROCUTICLE DEPOSITION



4. ECDYSIS



5. EXPANSION



6. DARKENING AND HARDENING



7. ENDOCUTICLE DEPOSITION



- At the onset of molting in response to the hormone ecdysone the epidermal cells become active
- They increase in size and often in cell numbers
- During molting process the epidermal cells separate from the old cuticle
- Simultaneously begin to secrete both a new epicuticle and procuticle
- This initial separation of old cuticle and epidermis is called *apolysis*
- The epidermal cells then secrete the molting fluid or exuvial fluid
- It contain inactive enzymes (chitinase and proteases)

- Those enzymes are capable of digesting endocuticle
- This fluid then changes to a gel
- Simultaneously, a new cuticle containing some layers of the epicuticle and the undifferentiated procuticle is produced
- This separate the molting gel from the epidermal cells
- Somehow (probably the pH change) activates the enzymes in the molting gel
- It begins to digest the old endocuticle
- 80% - 90% of the digested products will be absorbed by the epidermal cells

- Thus significant portion of the cuticle never entirely leaves the metabolic pool of insect
- Exocuticle and epicuticle are resistant to the action of the molting fluid
- It make up the portion of the integument that is shed at ecdysis
- As the old cuticle is digested space will be formed named *exuvial space*
- New cuticle will be deposited there
- First the cuticulin layer
- Then the inner epicuticle
- Finally the exocuticle and endocuticle

- The new cuticle is typically wrinkled beneath the old
- This indicate the greater surface area that it will have in the expanded insect – allowing the insect to grow
- The superficial layer will be laid down shortly after ecdysis
- This assure the water proofing of the newly emerged insect
- The cement layer is the last secreted which is formed after ecdysis by the dermal glands

- When the secretion of the new cuticle is complete the insect emerges leaving behind what remains of the old cuticle
- i.e. exuviae
- Following ecdysis an insect may eat the exuviae
- All the nutrients that may have been lost will be regain
- Sclerotization and melanization occur after ecdysis



THANK YOU!

